Tutorial - 01

A moving streaming Service wants to classify movies into Categories. Action or romance using a Knn classifier.

Input features

Action Scenes Count - no. of action sequences

Romance Scenes Count - no. of romantic Sequences.

Target variables

Genere (Action = A, Romance = R)

Movie ID	Action Scence	romance scence	Genre
1	15	2	A
2	1	12	R
3	18	1	A
4	2	14	R
5	14	3	A
6	3	11	R
7	16	2	R
8	2	13	R

Test dataset (with actual labels):

Test ID	Action Scene	romance Scene	Actual ge
T_1	10	5	A
T_2	4	10	R
T_3	17	2	A

Euclidean Formula:

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

(15,2)

$$= \sqrt{(10-15)^2 + (5-2)^2} = \sqrt{25+9} = \sqrt{34} = 5.83$$

(1,12)

$$= \sqrt{(10-1)^2 + (5-12)^2} = \sqrt{81+49} = \sqrt{130} = 11.4$$

(18,1)

$$= \sqrt{(10-18)^2 + (5-1)^2} = \sqrt{64+16} = \sqrt{80} = 8.94$$

(2,14)

$$= \sqrt{(10-2)^2 + (5-14)^2} = \sqrt{64+81} = \sqrt{145} = 12.04$$

(14,3)

$$= \sqrt{(10-14)^2 + (5-3)^2} = \sqrt{16+4} = \sqrt{20} = 4.47$$

(3,11)

$$= \sqrt{(10-3)^2 + (5-11)^2} = \sqrt{49+36} = \sqrt{85} = 9.21$$

(16,2)

$$= \sqrt{(10-16)^2 + (5-2)^2} = \sqrt{36+9} = \sqrt{45} = 6.70$$

(2,13)

$$= \sqrt{(10-2)^2 + (5-13)^2} = \sqrt{64+64} = \cancel{\sqrt{128}} = 11.31$$

Movie ID	distance	Genre
5	4.47	A
1	5.38	A
7	6.70	A

Classification of T_1

when $K=1$ Nearest Neighbours $(5,1,7)$ Genre - A

Prediction (A)

(x,y) $(4,10)$

$(15,2) = \sqrt{(4-15)^2+(10-2)^2} = \sqrt{121+64} = \sqrt{185} = 13.6$

$(1,12) = \sqrt{(4-1)^2+(10-12)^2} = \sqrt{9+4} = \sqrt{13} = 3.60$

$(18,1) = \sqrt{(4+8)^2+(10-1)^2} = \sqrt{196+81} = \sqrt{277} = 16.6$

$(2,14) = \sqrt{(4-2)^2+(10-14)^2} = \sqrt{4+16} = \sqrt{20} = 4.47$

$(14,3) = \sqrt{(4-14)^2+(10-3)^2} = \sqrt{100+49} = \sqrt{149} = 12.20$

$(3,11) = \sqrt{(4-3)^2+(10-11)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

$(16,2) = \sqrt{(4-16)^2+(10-2)^2} = \sqrt{144+64} = \sqrt{208} = 14.4$

$(2,13) = \sqrt{(4-2)^2+(10-13)^2} = \sqrt{4+9} = \sqrt{13} = 3.60$

movie id	distance	Genre
6	1.41	R
2	3.60	R
8	3.60	R

(17, 2)

$(15,2) = \sqrt{(17-15)^2 + (2-2)^2} = \sqrt{4+0} = 2$

$(1,12) = \sqrt{(17-1)^2 + (2-12)^2} = \sqrt{256+100} = 18.8$

$(18,1) = \sqrt{(17-18)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4$

$(2,14) = \sqrt{(17-2)^2 + (2-14)^2} = \sqrt{225+144} = \sqrt{369} = 19.2$

$(14,3) = \sqrt{(17-14)^2 + (2-3)^2} = \sqrt{9+1} = \sqrt{10} = 3.16$

$(3,11) = \sqrt{(17-3)^2 + (2-11)^2} = \sqrt{196+81} = \sqrt{277} = 16.6$

$(16,2) = \sqrt{(17-16)^2 + (2-2)^2} = \sqrt{1+0} = \sqrt{1} = 1$

$(2,13) = \sqrt{(17-2)^2 + (2-13)^2} = \sqrt{225+121} = \sqrt{346} = 18.60$

Movie ID	distance	Genre
17	1	A
3	1.4	A
1	2	A

Actual labels Vs predicted labels

Test ID	Actual	predicted (K=1)	Predicted (K=2)	predicted (K=3)
T_1	A	A	A	A
T_2	R	R	R	R
T_3	A	A	A	A

All predictions match with Actual labels

3. Accuracy = $\frac{\text{Correct predictions}}{\text{Total predictions}}$

$= \frac{3}{3} = 1 \times 100 = 100\%$ — Accuracy

4. Confusion Matrix.

Action → positive

Romance → Negative

	A	R
A	2	0
R	0	1

$T_P = 2$

$F_P = 0$

$F_N = 0$

$T_N = 1$

5. precision $= \frac{T_P}{(T_P + F_P)} = \frac{2}{2+0} = 1$

Precision = 100%.

F_1 Score

$$F_1 = \frac{2 \times (\text{precision} \times \text{Recall})}{(\text{precision} + \text{Recall})}$$

$$F_1 = \frac{2 \times (1 \times 1)}{1+1} = 1$$

F_1 Score = 100%